UGANDA MARTYRS SS NAMUGONGO (UMSSN) MATH SEMINAR 2026
PRINCIPAL MATHEMATICS
PAPER 2 SOLUTIONS

CONSTRUCT: NUMERIAL ANALYSIS

Item one
(a) **Trapezium rule and reinforcement decision**
The spacing is $h = 3\,m$.

x (m)	Endpoint ordinates y_0, y_6	Interior ordinates $y_1, \dots, y_5$
0	3.85	
3		5.20
6		5.50
9		5.20
12		3.85
15		3.00
18	0	
Sum	3.85	22.75

$$A \approx \frac{h}{2}[y_0 + y_6 + 2(y_1 + y_2 + y_3 + y_4 + y_5)]$$
$$\approx \frac{3}{2}[3.85 + 2(22.75)] = 74.025\ \text{m}^2.$$

Decision: Since $74.025\ \text{m}^2$ is less than $75\ \text{m}^2$, high-tensile steel reinforcement will not be required.

(b) Bounds, absolute error and budget

$$|\Delta w| = 27.56 \times \frac{4.2}{100} = 1.15752\ \text{m},$$
$$|\Delta L| = 41\frac{5}{100} = 2.05\ \text{m},$$
$$|\Delta c| = 6.25\frac{0.8}{100} = 0.05\ \text{m}.$$

Dimension	Minimum	Minimum
w	26.40248	28.71752
L	38.95	43.05
c	6.20	6.30

$A_{\max} = 28.71752(43.05 - 6.20) = 1058.240612\ \text{m}^2,$
$A_{\min} = 26.40248(38.95 - 6.30) = 862.040972\ \text{m}^2,$
$A \in [862.041625,\quad 1058.239875]\ \text{m}^2.$

absolute error $= \frac{1}{2}(1058.240612 - 862.040972) = 98.09982\ \text{m}^2$,
working value $= 27.56(41 - 6.25) = 957.71\ \text{m}^2$,
percentage error $= \frac{98.09982}{957.71} \times 100\% = 10.2432\%$.
Decision. The interval lies within the budget area range of 850 to $1100\ \text{m}^2$, so the budget is sufficient.

Item Two
Task (a)

True rate x (mL/min)	35	40
Displayed rate y (mL/min)	50	65

For display $y = 32$,
$\frac{x - 35}{32 - 50} = \frac{40 - 35}{65 - 50}$
$\Rightarrow \quad x = 29\ \text{mL/min}$.
For true rate $x = 38$,
$\frac{y - 65}{38 - 40} = \frac{50 - 65}{35 - 40}$
$\Rightarrow \quad y = 59\ \text{mL/min}$.
Decision. A predicted display of $59\ \text{mL/min}$ exceeds$55\ \text{mL/min}$; Tom must buy a new machine.

b) Reservoir volume by trapezium rule
$h = 4$ hours $= 4(60)(60) = 14400$ seconds.

t (hours)	Endpoint rates y_0, y_6	Interior rates $y_1, \ldots, y_5$
0	1.1542	
4		2.4891
8		3.8215
12		3.1048
16		2.9753
20		1.6340
24	0.9512	
Sum	2.1054	14.0247

$V \simeq \frac{1}{2}(14400)[2.1054 + 2(14.0247)] = 217114.56\ \text{m}^3$.
Decision. $217114.56 > 216500\ \text{m}^3$; the supply is stable and the auxiliary pumps need not be activated.

(c) Angular-error comparison
$\theta = 20° = 20 \times \frac{\pi}{180} = \frac{\pi}{9}$ rad,
$y = 3\cos(3\theta) = 1.5\ \text{kN}$,
$|\Delta y| \simeq |-9\sin(3\theta)\Delta\theta|$.

	Tool A	Tool B
Angular error	$0.6^\circ = \frac{\pi}{300}$ rad	$0.2^\circ = \frac{\pi}{900}$ rad
Estimated $\lvert\Delta y\rvert$	$\left\lvert 9\sin\left(\frac{\pi}{3}\right) \times \frac{\pi}{300}\right\rvert = 0.081621$ kN	$\left\lvert 9\sin\left(\frac{\pi}{3}\right) \times \frac{\pi}{900}\right\rvert = 0.027207$ kN
Relative error	$\frac{0.081621}{1.5} = 0.054414$	$\frac{0.027207}{1.5} = 0.018138$

Conclusion. Tool B has relative error $0.018138 < 0.05$ and should be purchased.

Item Four

a) Diameter $d = 3.40\,m,\ |\Delta d| = 0.005\,m$

Depth, $h = 1.20\,m,\ |\Delta h| = 0.005\,m$

Volume, $V = \pi r^2 h,\ r = \frac{d}{2}$

$V_{\max} = \pi\left(\frac{3.40 + 0.005}{2}\right)^2 (1.20 + 0.005)$

$\boldsymbol{V_{\max} = 3.492700031\pi\ m^3}$

$V_{\min} = \pi\left(\frac{3.40 - 0.005}{2}\right)^2 (1.20 - 0.005)$

$\boldsymbol{V_{\min}} = 3.443399969\pi\ m^3$

A.E $= \frac{1}{2}(3.492700031 - 3.443399969)\pi$

$= 0.024650031\pi\ m^3$

Working Value $= \pi\left(\frac{3.40}{2}\right)^2 \times 1.20$

$= 3.468\pi\ m^3$

Percentage Error

Percentage error $= \frac{0.024650031\pi}{3.468\pi} \times 100\%$

= 0.71079% < 10%

The C.E.O. should not buy another tank.

b) $T(x) = 2x - 3\cos\left(\frac{x}{2}\right)$

$T(1) = 2(1) - 3\cos\left(\frac{1}{2}\right) = -0.6327$

$T(2) = 2(2) - 3\cos\left(\frac{2}{2}\right) = 2.3791$

$T(1).T(2) = (-0.6327)(2.3791)$

$= -1.4994 < 0$

Since T (1) · T (2) < 0, there exists a root between 1 and 2.

Linear Interpolation – First Iteration

1	x_0	2
-0.6327	0	2.3791

$\frac{2.3791 - (-0.6327)}{(2-1)} = \frac{(0 - (-0.6327))}{(x_0 - 1)}$

$x_0 = 1.2101$

$T(1.2101) = 2(1.2101) - 3\cos\left(\frac{1.2101}{2}\right)$

$= -0.0473$

Linear Interpolation – Second Iteration

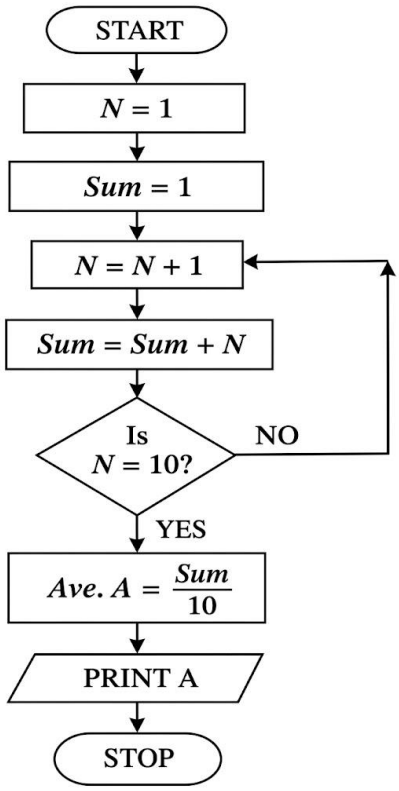
1.2101	x_1	2
-0.0473	0	2.3791

$$\frac{(2.3791-(-0.0473))}{(2-1.2101)} = \frac{0-(-0.0473)}{(x-1.2101)}$$

$x = 1.2255 < 1.5$

Decision: The refrigerator is considered safe for preservation**.**

Flowchart



Tracing the algorithm shown in the flowchart

N	S
1	1
2	3
3	6
4	10
5	15
6	21
7	28
8	36
9	45
10	55

Average $= \frac{55}{10} = 5.5 > 5$

Decision: The mixing tank is certified as stable.

Item Five

(a) **Total net cooling** effect using the trapezium rule

The rate at which the net cooling effect changes is . since the total cooling effect, T is required over $0 \leq x \leq \pi$ which is $T = \int_0^{\pi} \boldsymbol{x \cos x \, dx}$.

The monitoring system divides the interval into six equal sub-intervals. Therefore,

$h = \frac{(\pi - 0)}{6} = \frac{\pi}{6}$

$\boldsymbol{x}$	$\boldsymbol{f(x) = x \cos x}$	
0	0.00000	
$\frac{\pi}{6}$		0.45345
$\frac{\pi}{3}$		0.52360
$\frac{\pi}{2}$		0.00000
$\frac{2\pi}{3}$		−1.04720
$\frac{5\pi}{6}$		−2.26725
π	−3.14159	
SUM	**-3.14159**	**-2.33740**

Using the trapezium rule,

$T \approx \frac{1}{2} \times \frac{\pi}{6} [-3.14159 + 2(-2.3374)]$.

$T \approx -2.046326$.

Exact value $= \int_0^{\pi} \boldsymbol{x \cos x \, dx}$.

Using integration by parts, let $u = x$ and $\frac{dv}{dx} = \cos x$. Then $\frac{du}{dx} = 1$ and $v = \sin x$.

$$I = x \sin x \Big|_0^{\pi} - \int_0^{\pi} \sin x \, dx.$$

$$= x \sin x + \cos x \Big|_0^{\pi} = -2$$

Percentage error $= \frac{|E.V - A.V|}{|E.V|} \times 100 = \frac{|-2 - -2.046326|}{|-2|} \times 100\%$

$= 2.3163\% < 3\%$

Decision: The refrigeration system satisfies the manager's requirement.

(c) Refund to the sick passenger

Distance travelled (km)	10	20	30	40
Fares (shs)	11000	15000	18500	20000

Extract

30	33	40
18500	y	20000

$\frac{20000 - y}{40 - 33} = \frac{20000 - 18500}{40 - 30}$

$y = 18950$

Five percent of the fare paid $= 0.05 \times 2{,}0000 = Shs.1000$.

Value of unused part of the journey $= 2{,}0000 - 18950 = Shs.1050 \geq Shs.1000$

Decision: The passenger will receive a refund of Shs. 1050.

Distance travelled by the passenger with Shs. 16500

$\frac{x-20}{16500-15000} = \frac{30-20}{18500-15000}$

$x = 24.2857km < 25km$

20	x	30
15000	16500	18500

Decision: The passenger should not be allowed to board under the stated condition.

Range of possible gross income

Tilapia profit$(P_t) = Shs.180\ million$, $e_{P_t} = \frac{5}{100} \times 180 = Shs.9\ million.,.$

$Maximum\ Tilapia\ Profit = 180 + 9 = Shs.189\ million$

$Minimum\ Tilapia\ Profit = 180 - 9 = Shs.171 million$

Nile perch loss, $N_l =$ Shs. 100 million, $e_{N_l} = \frac{3}{100} \times 100 = Shs.3\ million$

$Maximum\ Nile\ perch\ Loss = 100 + 3 = Shs.103\ million$

$Maximum\ Nile\ perch\ Loss = 100 - 3 = Shs.97\ million.$

$\boldsymbol{Gross\ income = Profit - loss}$

Minimum gross income $= 171 - 103 = Shs.68\ million.$

Maximum gross income$= 189 - 97 = Shs.92\ million.$

Therefore, 68 million ≤ gross income ≤ 92 million.

<u>Tax as a percentage of gross income</u>

Tax percentage $= \frac{5}{Gross\ Income} \times 100$

Maximum tax percentage $= \frac{5}{68} \times 100 = 7.353$. (***Note this occurs at the minimum gross income***)

Minimum tax percentage $= \frac{5}{92} \times 100 = 5.435$ (***occurs at the maximum gross income***)

Therefore, 5.43% ≤ tax percentage ≤ 7.35%.

Since 7.35% < 7.5%, the condition is satisfied.

Decision: The financial condition is satisfied.

Item Six

a) $f(x) = 3x^2 + x - 5$

$f(1) = 3(1)^2 + 1 - 5 = -1$

$f(2) = 3(2)^2 + 2 - 5 = 9$

Since $\boldsymbol{f(1) \cdot f(2) = (-1)(9) = -9 < 0}$, the point lies in the interval [1, 2]

$x_0 = \frac{1+2}{2} = 1.5$

$f(1.5) = 3(1.5)^2 + 1.5 - 5 = 3.25$

x	**1**	$\boldsymbol{x_0}$	**1.5**
f(x)	−1	0	3.25

$$\frac{1.5-1}{3.25--1} = \frac{x_0 - 1}{0--1}$$

$$x_0 = 1.1176$$

$f(1.1176) = 3(1.1176)^2 + 1.1176 - 5 = -0.1353$

1.1176	$\boldsymbol{x_1}$	**1.5**
-0.1353	0	3.25

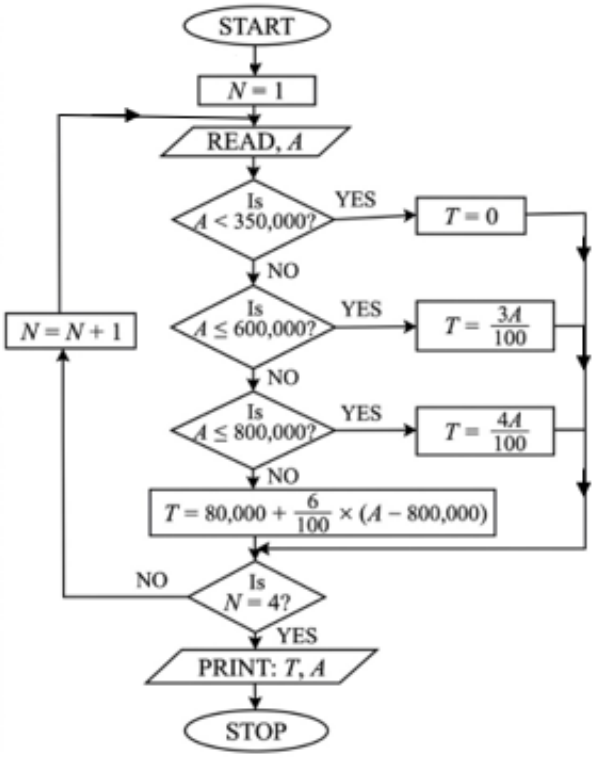
$$\frac{(1.5-1.1176)}{(3.25-0.1353-)} = \frac{(x_1 - 1.1176)}{(0-0.1353)}$$

$$x_1 = 1.13288\ (5\ d.ps.)$$

$1.13288 \times 100 = 113.288m < 150m,$
Decision: The engineer will **not** take fresh measurements.

(b) Given: $m = 48.6\,kg, \quad e_m = 0.05, \quad v = 120\,litres, \quad e_v = 0.5$
Concentration , $c = \frac{m}{v}$
$C_{max} = (\frac{(48.6 + 0.05)}{120 - 0.5)} = 0.407113$
$C_{min} = \frac{(48.6 - 0.05)}{120+0.5} = 0.402905$
A.E $= \frac{1}{2}(0.407113 - 0.402905) = 0.002104$
Exact Value, $C = \frac{48.6}{120} = 0.405$
Percentage error $= (\frac{0.002104}{0.405}) \times 100\%$
$= 0.51951\% < \mathbf{0.7}\%$
Decision: The engineer will consider the concentration firm.

c) **Flow chart**



N	A	T
1	250, 000	0
2	400, 000	12000
3	800, 000	32000
4	1,050,000	95000

Total tax paid $= 12000 + 32000 + 95000 = 139000 < 150{,}000$

Decision: The school should consider hiring more workers.

Item Seven

(a) **Given:** $\boldsymbol{y = xe^x}$, Interval: x = 0.5 km to x = 2.5 km

Step size: h = $\frac{(2.5-1)}{(5-1)}$ =0.5km

x (km)	y (km)	
0.5	0.8244	
1.0		2.7183
1.5		6.7225
2.0		14.7781
2.5	30.4562	
Sum	**31.2806**	**24.2189**

Using the Trapezium Rule:

Area ≈ ½ × 0.5 [31.2806 + 2 (24.2189)]

Area ≈ $\boldsymbol{19.930\ km^2}$, 19 < 19.930 < 20

Decision: Estimated area lies in the Substation Standard area**.**

(b) **Exact area=** $\boldsymbol{\int_{0.5}^{2.5} x\, e^x dx}$

Integration by parts:

Let $u = x \Rightarrow \frac{du}{dx} = 1$

$\frac{dv}{dx} = e^x \Rightarrow \Rightarrow v = e^x$

$\int x\, e^x\, dx = [x\, e^x] - \int e^x\, dx$

$= [x\, e^x]_{0.5}^{2.5} - [e^x]_{0.5}^{2.5}$

$= (2.5\, e^{2.5} - 0.5\, e^{0.5}) - (e^{2.5} - e^{0.5})$

$= 2.5\, e^{2.5} - 0.5\, e^{0.5} - e^{2.5} + e^{0.5}$

Exact area = $\boldsymbol{19.098\ km^2}$

Percentage error = $\frac{|19.098-19.930|}{19.098}$× 100%

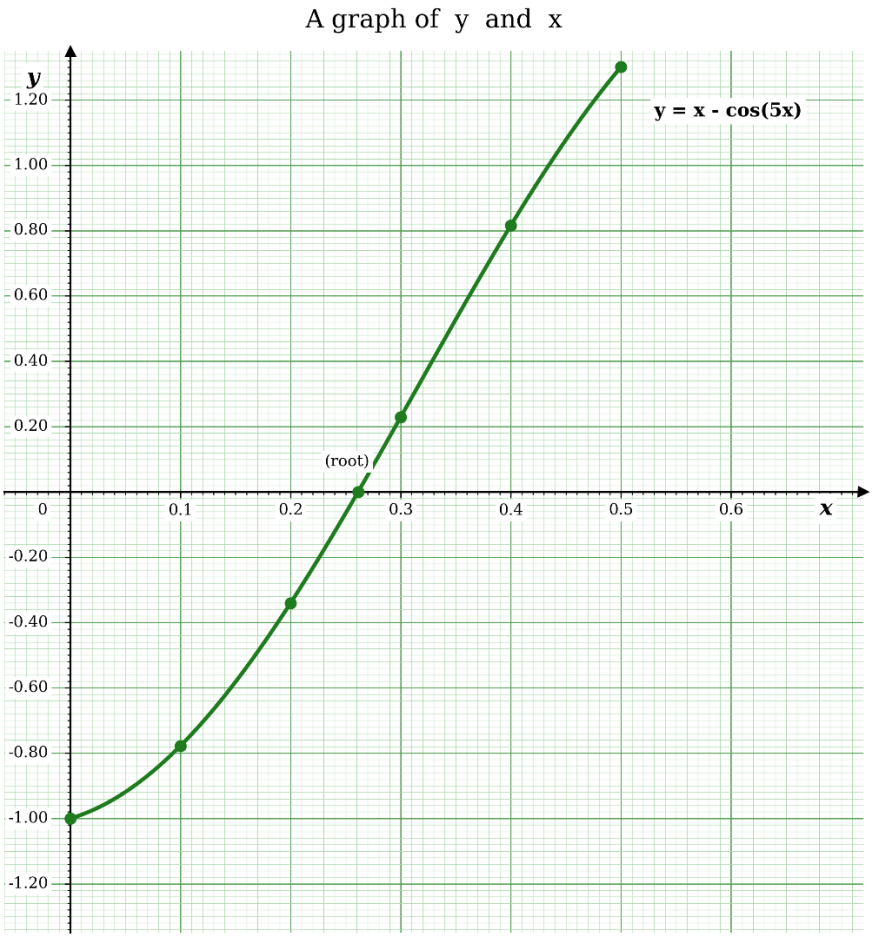
= **4.3565%**, 4 < 4.3565 < 5

Decision: The company can approve the construction**.**

(c) $y = x - cos\,(5x)$

Table of values:

x	0	0.1	0.2	0.3	0.4	0.5
y	−1	−0.78	−0.34	0.23	0.82	1.30



Initial approximation: $x_0 = 0.27$ (from the graph)

$At\ y = 0 \Rightarrow\ x - cos\ (5x) = 0$

$f(x) = x - cos(5x)$

$f'(x) = 1 + 5\ sin(5x)$

$x_{n+1} = x_n - \frac{f(x_n))}{f\prime(x_n)}$

$x_{n+1} = x_n - \frac{(x_n - cos(5x_n))}{(1 + 5\ sin(5x)}$

$x_{n+1} = \frac{[5\ x_n\ sin(5x_n) + cos(5x_n)]}{1 + 5\ sin(5x)}$ $\quad for\ n = 0,1,2, \ldots$

Iterations:

$X_0 = 0.27$

$X_1 = \frac{[5(0.27)\ sin(5 \times 0.27) + cos(5 \times 0.27)]}{1 + 5\ sin(5 \times 0.27)} = 0.26133$

$|x_1 - x_0| = |0.26133 - 0.27| = 0.00867 > 0.00001$

$X_2 = \frac{[5(0.26133)\ sin(5 \times 0.26133) + cos(5 \times 0.26133)]}{[1 + 5\ sin(5 \times 0.26133)]} = 0.26129$

$|x_2 - x_1| = |0.26129 - 0.26133| = 0.00004 > 0.00001$

$X_3 = \frac{[5(0.26129)sin(5 \times 0.26129) + cos(5 \times 0.26129)]}{[1 + 5\ sin(5 \times 0.26133)]} = 0.26129$

$|x_3 - x_2| = |0.26129 - 0.26129| = 0.00000 < 0.00001$

The root $= 0.2613\ Mv < 0.3000\ Mv.$

Decision: The grid will not remain stable, balanced and capable of preventing blackouts

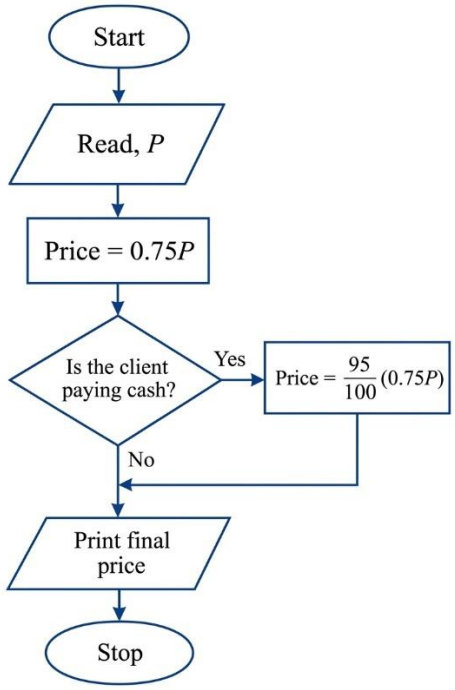
Item Eight

Discount Pricing Flowchart

Let P denote the marked price

Payment type	Final price
Credit	$0.75P$
Cash	$\frac{95}{100}(0.75P)$

Flowchart



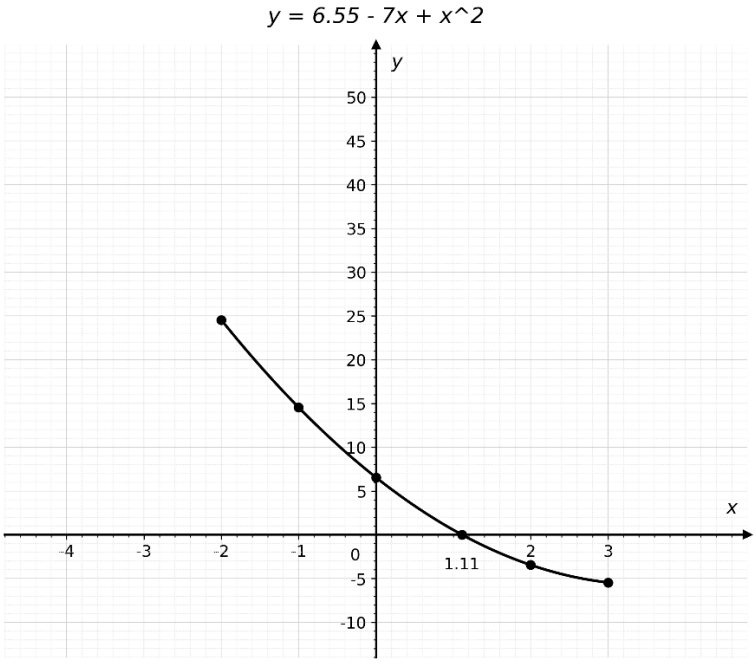
Dry run

Marked price	Payment method	Final price
Shs 12,500,000	Credit	Shs 9,375,000
Shs 34,000,000	Cash	Shs 24,225,000
TOTAL		**Shs 33,600,000**

Decision: The buyer should continue with the purchase based on the calculated final price.

Given: $y = x^2 - 7x + 6.55$.

x	-4	-3	-2	-1	0	1	2	3
y	50.55	36.55	24.55	14.55	6.55	0.55	-3.45	-5.45



$x_o = 1.11 > 1.0$

Decision: The dealer must not give a bigger discount for this specific plot.

Estimation of Area Using the Trapezoidal Rule

Given: $y = x^2 - 7x + 6.55$, with the interval from $x = 2\ to\ x = 3$.

$\frac{h = (3-2)}{6} = \frac{1}{6}$

$\boldsymbol{x}$	$\boldsymbol{y} = x^2 - 7x + 6.55$	
2	-3.4500	
$\frac{13}{6}$		-3.9222
$\frac{7}{3}$		-4.3389
$\frac{5}{2}$		-4.7000
$\frac{8}{3}$		-5.0056
$\frac{17}{6}$		-5.2556
3	-5.4500	
SUM	**-8.9000**	**-23.2223**

Apply the trapezoidal rule

$A \approx \frac{1}{2} \times \frac{1}{6} [-8.9000 + 2(-23.2223)]$

$A \approx -4.61205$

Area enclosed $\approx |-4.612| \approx \boldsymbol{4.61\ square\ units} > 1.25$ square units**.**

Decision: The situation would attract the specified environmental penalty.

CONSTRUCT: PROBABILITY AND STATSTICS

Item One

$$F(x) = \begin{cases} 0, & x \leq 0, \\ k\dfrac{x^2}{2}, & 0 < x < 2, \\ k(6x - x^2 - 6), & 2 < x < 3, \\ 1, & x \geq 3. \end{cases}$$

$F(3) = 1$
$k[6(3) - 3^2 - 6] = 1$
$3k = 1$
$k = \frac{1}{3}$

$$F(x) = \begin{cases} 0, & x \leq 0, \\ \dfrac{1}{6}x^2, & 0 < x < 2, \\ \dfrac{1}{3}(6x - x^2 - 6), & 2 < x < 3, \\ 1, & x \geq 3. \end{cases}$$

For median, $F(m) = 0.5$
For $x = 2$, $F(2) = \frac{1}{6}(2^2) = \frac{4}{6} > 0.5$
Since $\frac{4}{6} > 0.5$, the median lies within the interval $0 < x < 2$
$F(m) = \frac{1}{6}(m^2) = 0.5$
$\mathrm{m}^2 = 3$
$m = \sqrt{3} \approx 1.732\ hours$
For $0 < x < 2$: $f(x) = \frac{d}{dx}\left(\frac{1}{6}x^2\right) = \frac{1}{3}x$
For $2 < x < 3$: $f(x) = \frac{d}{dx}\left(2x - \frac{1}{3}x^2 - 2\right) = 2 - \frac{2}{3}x$

$$f(x) = \begin{cases} \dfrac{1}{3}x, & 0 < x < 2, \\ 2 - \dfrac{2}{3}x, & 2 < x < 3, \\ 0, & x > 3. \end{cases}$$

$$E(X) = \int_0^2 x\left(\frac{1}{3}x\right)dx + \int_2^3 x\left(2 - \frac{2}{3}x\right)dx$$
$$\left[\frac{1}{9}x^3\right]_0^2 + \left[x^2 - \frac{2}{9}x^3\right]_2^3 = \left(\frac{1}{9}[2]^3 - \frac{1}{9}(0)^3\right) + \left[\left(3^2 - \frac{2}{9}[3]^3\right) - \left(2^2 - \frac{2}{9}[2]^3\right)\right]$$
$\frac{8}{9} + \frac{7}{9} = \frac{5}{3} = 1.66667\ hours$
Since both statistical criteria are completely satisfied, the school management will retain the current three-hour assessment session.

Science enrolment and conditional probability
The handwritten values are $P(M) = 0.55$, $P(F) = 0.45$, $P(S/M) = 0.30$ and $P(S/F) = 0.05$.

$P(S) = \mathrm{P}(\mathrm{F} \cap \mathrm{S}) + \mathrm{P}(\mathrm{M} \cap \mathrm{S})$
$P(S) = P(F) \times P\left(S/_F\right) + P(M) \times P\left(S/_M\right)$
$P(S) = 0.55(0.30) + 0.45(0.05) = \frac{3}{16},$
$P(S^c) = 1 - \frac{3}{16}.$
$P(S^c) = \frac{13}{16}.$
$P\left(S^c/_F\right) = 1 - P\left(S/_F\right) = 1 - 0.05.$
$\left(S^c/_F\right) = 0.95$
$P\left(F/_{S^c}\right) = \frac{\mathrm{P}(\mathrm{F} \cap S^c)}{S^c} = \frac{0.45 \times 0.95}{\frac{13}{16}} = \frac{171}{325}$
Since $\frac{171}{325} \geq 0.5$, the school will give additional attention to supporting non-science subjects.

Spearman ranks for two tests

R_x	R_y	$d = R_x - R_y$	d^2
5	6	-1	1
3	1	2	4
2	4	-2	4
6	5	1	1
4	2	2	4
1	3	-2	4
$\sum d^2$			18

$\rho = 1 - \frac{6(18)}{6(6^2 - 1)} = 0.4857.$
Since $0.4857 < 0.70$, the school should not regard the two tests as sufficiently consistent for use together in assessing students.

Item Two
Uniform waiting time
For $X \sim R(0{,}20)$ minutes, $f(x) = \frac{1}{20}$ on $[0, 20]$. Thus
$f(x) = \begin{cases} \frac{1}{20} & 0 < x < 20 \\ 0 & elsewhere \end{cases}$
$P(X > 15) = \int_{15}^{20} \frac{1}{20}\, dx = \frac{5}{20} = 0.25 > 0.20,$
$E(X) = \frac{0 + 20}{2} = 10$ minutes.

Since the calculated probability of missing the opening time $0.25 \geq 0.20$, Alex will cancel the current contract.

Discount decision

$$f(x) = \begin{cases} \frac{3}{16}x^2, & 0 \leq x \leq 2, \\ \frac{3}{16}(4-x)^2, & 2 \leq x \leq 4, \\ 0, & \text{otherwise.} \end{cases}$$

$$P(1 \leq X \leq 3) = \int_1^2 \frac{3x^2}{16}\,dx + \int_2^3 \frac{3(4-x)^2}{16}\,dx$$

$$= \left[\frac{1}{16}x^2\right]_1^2 + \left[\frac{1}{16}(4-x)^3\right]_2^3$$

$$= \left[\left(\frac{1}{16}(2)^2\right) - \left(\frac{1}{16}(1)^2\right)\right] + \left[\left(\frac{1}{16}(4-3)^3\right) - \frac{1}{16}(4-2)^3\right]$$

$$= \frac{7}{16} + \frac{7}{16}$$

$$= \frac{7}{8} = 0.875.$$

Decision: Since $0.875 > 0.8$, Alex does not need to issue apology discount coupons

Mean and SD

By observing the symmetry of $f(x)$ around $x = 2$:

$E(X) = 2.0$ mintes

$$E(X^2) = \int_0^2 \frac{3x^4}{16}\,dx + \int_2^4 \frac{3x^2(4-x)^2}{16}\,dx$$

$$E(X^2) = \left[\left(\frac{3x^5}{80}\right)\right]_0^2 + \left[16x^3 - 6x^4 - \frac{3}{5}x^5\right]_2^4$$

$$\left[\left(\frac{3(2)^5}{80}\right) - \left(\frac{3(0)^5}{80}\right)\right] + \left[\left(16(4)^3 - 6(4)^4 - \frac{3}{5}(4)^5\right) - \left(16(2)^3 - 6(2)^4 - \frac{3}{5}(2)^5\right)\right]$$

$$E(X^2) = \frac{22}{5}$$

$$\sigma = \sqrt{\frac{22}{5} - 2^2}$$

$\sigma = 0.6325$ minutes

Conclusion: Since $0.6325 > 0.5$, Alex must schedule a mandatory speed-training workshop for the staff.

Spearman ranks for the campaign

R_x	R_y	d	d^2
1	2	−1	1
6	6	0	0
3	3	0	0
7	7	0	0
2	1	1	1
5	5	0	0
4	4	0	0
$\sum d^2$			2

$$\rho = 1 - \frac{6(2)}{7(7^2 - 1)}$$
$$= 0.9643$$

Since 0.9643 > 0.85, there is a very strong positive rank correlation showing that increasing online flyers significantly relates to higher revenue. Alex should approve the budget for the online campaign.

Item Three

Marks (%) — Frequency Table and Mode

Marks	**f**	C	$f.d$	cf
10–20	10	10	1.0	10
20–25	16	5	3.2	26
25–40	12	15	0.8	38
40–60	10	20	0.5	48
60–70	20	10	2.0	68
70–95	24	25	0.96	92
95–100	8	5	1.6	100

Modal class $= (20 - 25)$

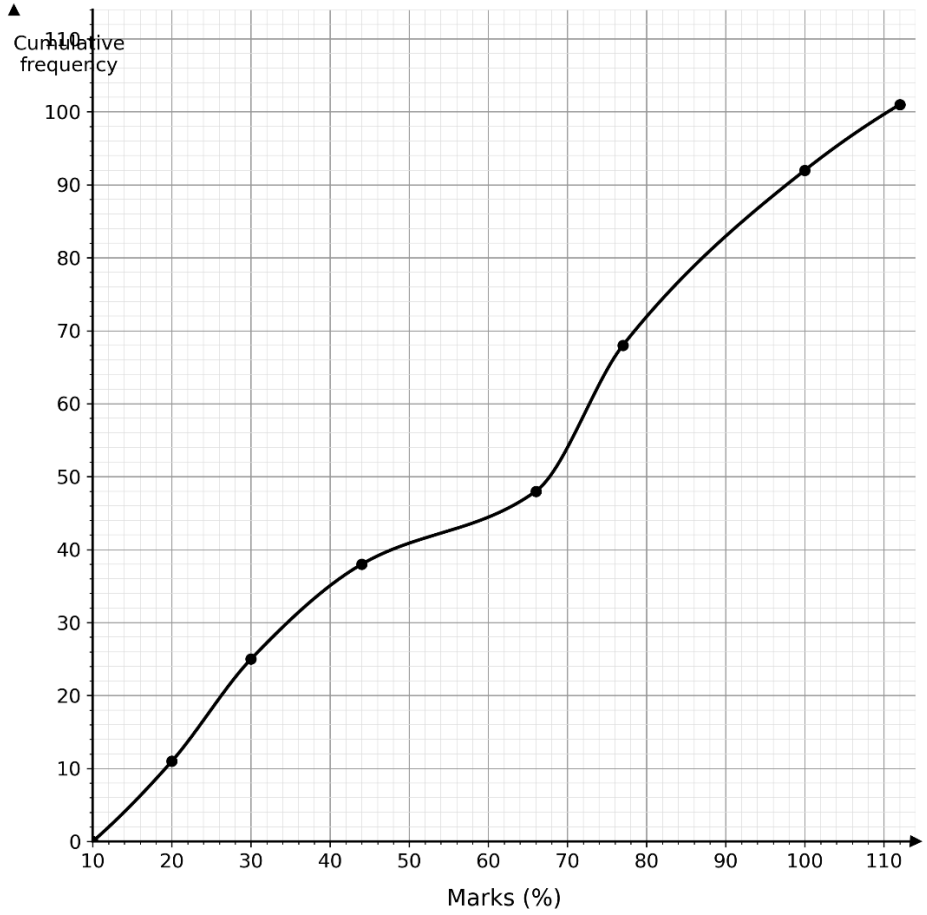
Mode $= L_1 + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) \times c$

$Mode = 20 + \left(\frac{2.2}{2.2 + 2.4}\right) \times 5 = \mathbf{22.39}$

Number of students $= 100 - 16 = 84 > 70$

Decision: The mathematics teacher will submit the Marks of the learners

Cumulative Frequency Graph



Probability calculation

$p = \frac{2}{6} = \frac{1}{3}$, for a 2 showing up. $q = \frac{2}{3}$,

X ~ B (8, $\frac{1}{3}$)

$$P(X \geq 4) = 1 - P(X \leq 3)$$
$$= 1 - [P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)]$$
$$= 1 - \left[{}^8C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^8 + {}^8C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^7 + {}^8C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^6 + {}^8C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^5\right]$$
$$= 1 - (0.0390 + 0.1561 + 0.2731 + 0.2731)$$
$$P(X \geq 4) = 0.2587$$

For a 4 showing up, $p = \frac{1}{6}$ and $q = \frac{5}{6}$.

Let Y ~ B (8, $\frac{1}{6}$)

$$P(Y \leq 5) = 1 - P(Y \geq 6)$$
$$= 1 - \left[{}^8C_6 \left(\frac{1}{6}\right)^6 \left(\frac{5}{6}\right)^2 + {}^8C_7 \left(\frac{1}{6}\right)^7 \left(\frac{5}{6}\right)^1 + {}^8C_8 \left(\frac{1}{6}\right)^8 \left(\frac{5}{6}\right)^0\right]$$
$$= 1 - (0.000416 + 0.000023 + 0.000006)$$
$$= 0.9999 > 0.2587$$

∴ The student will join the Maths Contest Club.

Probability of hitting

Let the probability of each student hitting be P (A) and P (B) respectively.

$$P(A \cap B) = 0.25$$
$$P(A) \cdot P(B) = 0.25$$
$$P(A \cup B) = 0.75$$
$$P(A) + P(B) - P(A) \cdot P(B) = 0.75$$

Let P (A) be a and P (B) be b.

$ab = 0.25 \ldots\ldots\ldots\ldots\ldots\ldots (1)$

$a + b - 0.25 = 0.75$

$a + b = 1.0$

$a = 1 - b \ldots\ldots\ldots\ldots\ldots\ldots (2)$

Substitute (2) into (1).

$(1 - b)b = 0.25$

$b - b^2 = 0.25$

$b^2 - b + 0.25 = 0$

$$b = \frac{[-(-1) \pm \sqrt{((-1)^2 - 4(1)(0.25))}}{2 \times 1}$$

$b = 0.5$.

From (2), $a = 1 - 0.5 = 0.5$.

Decision: Since P (A) = P (B) = 0.5 > 0.45, both students qualified for the national archery team.

Continuous probability distribution

$$F(x) = \begin{cases} 0; & x < 0 \\ ax; & 0 \le x \le 1 \\ b\tan^{-1}x & 1 < x \le \sqrt{3} \\ 1 & x \ge \sqrt{3} \end{cases}$$

$b\,tan^{-1}(\sqrt{3}) = 1$

$\left(\frac{\pi}{3}\right) b = 1$

$$b = \frac{3}{\pi}$$

$a(1) = b\,tan^{-1}(1)$

$a = \left(\frac{\pi}{4}\right) b\,, a = \left(\frac{\pi}{4}\right)\left(\frac{3}{\pi}\right), a = ¾$

Range	$\boldsymbol{F(x)}$	$\boldsymbol{f(x) = \frac{dF(x)}{dx}}$
$x < 0$	0	0
$0 \le x \le 1$	$\frac{3}{4}x$	$\frac{3}{4}$
$1 < x \le \sqrt{3}$	$\frac{3}{\pi}tan^{-1}x$	$\frac{3}{\pi(1+x^2)}$
$x > \sqrt{3}$	1	0

$$f(x) = \begin{cases} \frac{3}{4} & 0 \le x \le 1 \\ \frac{3}{\pi(1+x^2)} & 1 < x \le \sqrt{3} \\ 0 & elsewhere \end{cases}$$

Expectation and probability

$$E(X) = \int_0^1 \frac{3}{4}x\ dx + \int_1^{\sqrt{3}} \frac{3x}{\pi(1+x^2)}\ dx$$

$$= \frac{3}{4}\left[\left(\frac{x^2}{2}\right)\right]_0^1 + \left[\left(\frac{3}{2\pi}\right)\ln(1+x^2)\right]_1^{\sqrt{3}}$$

$= \frac{3}{8} + (\frac{3}{2\pi})[\ln 4 - \ln 2]$

$= \frac{3}{8} + (\frac{3}{2\pi}))\ln 2$

E(X) ≈ 0.7060 hours.

P (|X − 1| < 0.5) = P (−0.5 < (X − 1) < 0.5)

= P (0.5 < X < 1.5)

$= F(1.5) - F(0.5)$

$= (\frac{3}{\pi})\tan^{-1}(1.5) - (\frac{3}{4})(0.5)$

= 0.9385 − 0.375

= 0.5635

P (|X − 1| < 0.5) ≈ 0.5635 < 0.7060 hours

Decision: The geography teacher will choose the alternative route.

Item Four

Height (cm)	Frequency (f)	Midpoint (x)	fx	Cumulative frequency Cf
140–150	7	145	1015	7
150–160	6	155	930	13
160–170	15	165	2475	28
170–180	12	175	2100	40
180–190	7	185	1295	47
190–200	13	195	2535	60
	$\Sigma f = 60$		$\Sigma fx = 10350$	

Mean

$\bar{x} = \frac{\Sigma fx}{\Sigma f}$

$= \frac{10350}{60}$

= 172.5 cm.

Median class and Median

Median class: $\frac{1}{2}$ N = $\frac{1}{2}$ (60) = 30th value.

The 30th value lies in the class 170 – 180 (since the cumulative frequency just exceeds 30 in this class).

$\text{Median} = L_i + \left(\frac{(\frac{N}{2} - \text{Cfb})}{f_m}\right)c$

$= 170 + \left(\frac{(30 - 28)}{12}\right)10$

$= 170 + (\frac{1}{12})\ 10$

= 171.6667 cm.

Difference $= 172.5 - 171.6667$

$= 0.83\ cm < 2\ cm.$

∴ The data will be approved for selection.

$P(T) + 3P(T) = 1$

$4P(T) = 1$

P(T) = $\frac{1}{4}$ and P(H) = $\frac{3}{4}$

Let X ~ B(10, $\frac{3}{4}$).

$$P(X \leq 6) = 1 - \left[\binom{10}{7}\left(\frac{3}{4}\right)^7\left(\frac{1}{4}\right)^3 + \binom{10}{8}\left(\frac{3}{4}\right)^8\left(\frac{1}{4}\right)^2 + \binom{10}{9}\left(\frac{3}{4}\right)^9\left(\frac{1}{4}\right)^1 + \binom{10}{10}\left(\frac{3}{4}\right)^{10}\left(\frac{1}{4}\right)^0 \right]$$

= 1 − (0.2503 + 0.2816 + 0.1277 + 0.0563).

= 0.2241 < 0.5.

∴ The student will not qualify to join the team.

2. Let P(A) and P(B) be the probability of John and James shooting a goal according to a point respectively.

P(A) = 0.7, P(B) = 0.6.

P($\bar{A}$) = 0.3, P($\bar{B}$) = 0.4.

$$P(James\ win) = P(\bar{A}_1) + P(A_1 \cap B_1 \cap \bar{A}_2) + P(A_1 \cap B_1 \cap A_2 \cap B_2 \cap \bar{A}_3) + \cdots$$

$$= 0.3 + 0.7 \times 0.6 \times 0.3 + 0.7 \times 0.6 \times 0.7 \times 0.6 \times 0.3 + \cdots$$

$$= 0.3 + 0.126 + 0.05292 + \cdots$$

from $S\infty = \frac{a}{1-r}$

$$= \frac{0.3}{1 - 0.42}$$

$$P(James\ win) = 0.5172.$$

P(John win in 3rd turn) $= P(A_1 \cap B_1 \cap A_2 \cap B_2 \cap A_3 \cap B_3')$

$$= 0.7 \times 0.6 \times 0.7 \times 0.6 \times 0.7 \times 0.4$$

$$= 0.0494$$

$$= 0.0494 < 0.5172.$$

∴ James will qualify for the regional squad.

Let X ~ U (0.5, 2.5).

$$\frac{1}{b-a} = \frac{1}{2.5-0.5} = 0.5$$

$$f(x) = \begin{cases} 0.5, & 0.5 \leq x \leq 2.5 \\ 0, & elsewhere. \end{cases}$$

E(X) $= \frac{a+b}{2}$

$= \frac{2.5+0.5}{2}$

$= \frac{3}{2}$

= 1.5 kg.

P(X < E(X)) = P(X < 1.5)

$$= \int_{0.5}^{1.5} 0.5dx$$

$$= 0.5x\Big|_{0.5}^{1.5}$$

= 0.5[1.5 − 0.5]

= 0.5.

P(X < 1.5) = 0.5 > 0.3

∴ The batch will be classified as Grade A.

Item Five

(a)

R_x	R_y	d	d^2
1	3	-2	4
2	2	0	0
3	4	-1	1
4	6	-2	4
5	1	4	16
6	5	1	1
7	7	0	0
8	8	0	0
9	9	0	0
10	10	0	0
			$\sum \boldsymbol{d^2 = 26}$

$\rho = 1 - \frac{6\sum d^2}{n(n^2-1)}$

$= 1 - \frac{6(26)}{10(10^2-1)}$

$= 0.8424 > 0.70$

The committee will consider height useful as an indicator of throwing performance

(b) P (P) = 0.6 P (P') = 0.4

P(C) = 0.75 P(C') = 0.25

P (M) = 0.80 P (M') = 0.20

Let T be the event of passing at least two subjects.

P (Otim fails Chemistry given that he passes at least two subjects).

$P\left(C^1/_T\right) = \frac{P(C^1 \cap T)}{P(T)}$

$P(C^1 n\, T) = (0.6 \times 0.8 \times 0.25)$

$= 0.12$

$P(T) = P(P \cap C \cap M^1) + P(P \cap C^1 \cap M) + P(P^1 \cap C \cap M) + P(P \cap C \cap M)$

$= (0.6 \times 0.75 \times 0.2) + (0.6 \times 0.25 \times 0.8) + (0.4 \times 0.75 \times 0.80) + (0.6 \times 0.75 \times 0.80)$

$= \mathbf{0.81}$

$P\left(C^1/_T\right) = \frac{0.12}{0.81} = \frac{4}{27}$

$= 0.1481 < 0.20,$

Decision: The committee will not recommend additional Chemistry support.

(c) Since $n > 2500$, we introduce Normal approximation to binomial, $X \sim N(\mu, \sigma^2)$

$\mu = np = 2500 \times 0.2 = 500$

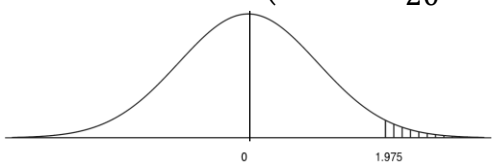
$\sigma^2 = npq = 2500 \times 0.2 \times 0.8 = 400$

Hence $\sigma = \sqrt{400} = 20$

$\therefore \boldsymbol{X \sim N(500, 20^2)}$

Probability of at least 540 visiting team supporters:

$P(X \geq 540) = P\left(Z > \frac{539.5 - 500}{20}\right)$



$P(Z > 1.975) = 0.5 - P(0 < Z < 1.975)$

$= 0.5 - 0.4759$

$= 0.0241 < 0.05$

Decision: The security committee will not make additional preparations.

(d) $n = 49,\ \bar{X} = 49,\ s = 2$ (for the sample)

From $\hat{\delta} = \sqrt{\frac{nS^2}{n-1}} = \sqrt{\frac{49\ x\ 4}{48}} = 2.0207$

$\alpha = 0.8$, $\frac{\alpha}{2} = 0.4$ and $Z_{\frac{\alpha}{2}} = 1.282$

C.I = $\bar{X} \pm Z_{\frac{\alpha}{2}} . \frac{\hat{\delta}}{\sqrt{n}}$

Upper limit = $49 + 1.282 \times \frac{2.0207}{\sqrt{49}}$

Upper limit = 49.3701 minutes.

Item Six

Part (i): Frequency table

x	f	fx	x^2	fx^2
470	1	470	220900	220900
475	1	475	225625	225625
480	1	480	230400	230400
485	2	970	235225	470450
490	2	980	240100	480200
495	2	990	245025	490050
500	2	1000	250000	500000
505	2	1010	255025	510050
510	2	1020	260100	520200
515	1	515	265225	265225
520	2	1040	270400	540800
525	1	525	275625	275625
530	1	530	280900	280900
SUM	**20**	**10005**		**5010425**

$Mean = \frac{\Sigma fx}{\Sigma f} = \frac{10005}{20} = \mathbf{500.25g} > 500\ g$

$SD = \sqrt{\frac{\Sigma f(x^2)}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2}$

$= \sqrt{\left(\frac{5010425}{20}\right) - \left(\frac{10005}{20}\right)^2}$

$= \mathbf{16.47g} > 10$ g

Decision: The production process will be suspended.

Part (ii)

Number of defectives = 2 out of 20.

p = $\frac{2}{20}$ = 0.1, q = 1 − 0.1 = 0.9

Let X be the number of defective booklets among the 4 selected booklets.

X ~ B (4, 0.1)

$P(X \geq 3) = P(X = 3) + P(X = 4)$
$P(X = 3) = {}^4C_3(0.1)^3(0.9)$
$= 4(0.001)(0.9) = 0.0036$
$P(X = 4) = {}^4C_4(0.1)^4(0.9)^0$
$= 1(0.0001)(1) = 0.0001$
$\Rightarrow P(X \geq 3) = 0.0036 + 0.0001 = 0.0037$
$= 0.37\% < 1\%,$
Since $0.37\% < 1\%$, the occurrence of defective booklets is not considered usual.
Decision: The treasury department should NOT prepare the 50% refund**.**

Part (iii)
From Part (i), use μ = 500.25 g and σ = 16.47 g.
Sample size: n = 36. We require $P(\bar{X} < 495)$.
Standard error = $\frac{\sigma}{\sqrt{n}} = \frac{16.47}{\sqrt{36}} = \frac{16.47}{6} = 2.745$ g
$Z = \frac{(\bar{X} - \mu)}{\sigma_{\bar{x}}} = \frac{495 - 500.25}{2.745} = -1.913$
$P\ (Z < -1.913) = 0.0281(tab)$
Therefore, P ($\bar{X}$ < 495) = 0.0281 = 2.81% < 5%.
Decision: UNBS should NOT take the stated regulatory action**.**

Item Seven

Part (1)

Age (x)	f	fx	fx^2
22	6	132	2904
23	8	184	4232
24	12	288	6912
25	10	250	6250
26	9	234	6084
27	5	135	3645
SUM	**50**	**1223**	**30027**

$$\sigma = \sqrt{\left(\frac{30027}{50}\right) - \left(\frac{1223}{50}\right)^2}$$

$= 1.4995 < 1.5\ years.$

Therefore, additional workers will not be recruited.

Let X be the number of packets sold.

x	5000	6000	7000	8000
$P(X = x)$	0.15	0.30	0.40	0.15
$xP(X = x)$	750	1800	2800	1200

$E(X) = 750 + 1800 + 2800 + 1200 = 6650$
Let the profit be Y, Y = Revenue − Cost.
$Y = 1500X - 7{,}000{,}000.$
$E(Y) = 1500E(X) - 7{,}000{,}000.$
$= (1500 \times 6550) - 7{,}000{,}000.$
$= 2{,}825{,}000 > 2{,}500{,}000.$
Decision: Therefore, the company will expand its operation.

Part 2

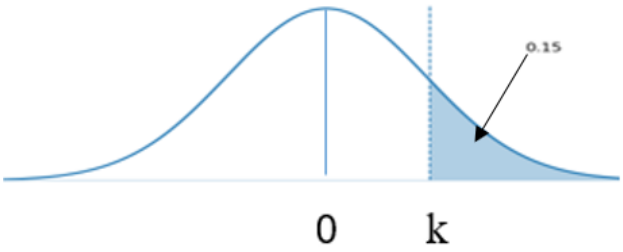
Let the mass of the detergent be X.

$X \sim N(\mu, \sigma^2)$

$P(X > 2.10) = 0.15.$

$P\left(Z > \frac{2.10 - \mu}{\sigma}\right) = 0.15.$

Let $\frac{(2.10 - \mu)}{\sigma} = k, \Rightarrow \mathrm{P(Z > k)} = 0.15.$



$P\ (0 < Z < k) = 0.5 - 0.15 = 0.35.$

k = 1.036 (from table)

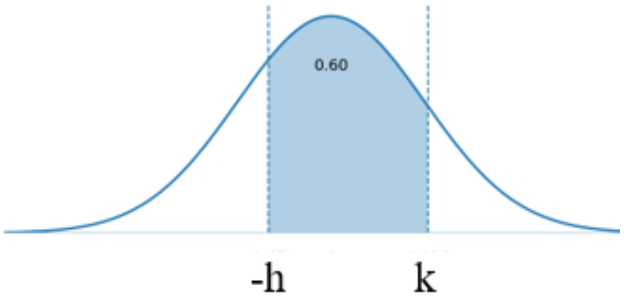
$\frac{(2.10 - \mu)}{\sigma} = 1.036$

$\Rightarrow 2.10 = \mu + 1.036\sigma \ldots\ldots (1)$

Given $P(1.90 < X < 2.10) = 0.60.$

$P(\frac{1.90 - \mu}{\sigma} < Z < \frac{2.10 - \mu}{\sigma}) = 0.60.$s

Let $\frac{1.90 - \mu}{h} = h \Rightarrow P(h < Z < k) = 0.60.$



The area between h and k is 0.60.

Thus, $P\ (-h < Z < 0) = 0.25.$

$-h = 0.674.$

$h = -0.674.$

$\frac{(1.90 - \mu)}{\sigma} = -0.674.$

Therefore, $1.90 = \mu - 0.674\sigma. \ldots\ldots (2)$

Solving equations (1) and (2):

$\mu + 1.036\sigma = 2.10$

$\mu - 0.674\sigma = 1.90$

Subtracting:

$(1.036 + 0.674)\sigma = 2.10 - 1.90$

$1.710\sigma = 0.20, \sigma = 0.1170\ kg.$

$\mu = 2.10 - (1.036 \times 0.1170)$

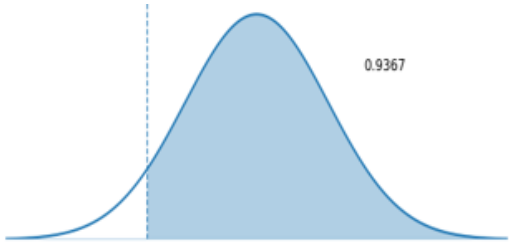
$\mu = 1.9788\ kg.$

Therefore, X ~ N (1.9788, 0.1170^2).

Now find $P(X > 1.80)$.

$P(X > 1.80) = P\left(Z > \frac{1.80 - 1.9788}{0.1170}\right)$

$= P(Z > -1.528)$.



$P(Z > -1.528) = P(-1.528 < z < 0) + 0.5$

$= 0.4367 + 0.5$

$= 0.9367 < 0.97$

Decision: Therefore, the company will not continue to use the filling machine.

Construct: Mechanics

Item One

(a)

For collision; $\begin{pmatrix}2t\\4t\end{pmatrix} = \begin{pmatrix}8\\8+2t\end{pmatrix}$

$\tilde{\boldsymbol{\imath}} : 2t = 8$

$t = 4\ seconds$

$\tilde{\boldsymbol{\jmath}} : 4t = 8 + 2t$

$t = 4\ seconds$

Since both components yield the exact same time, $t = 4\ seconds$, a collision is possible.

New position vector of A, at any time, t:

$\tilde{\mathbf{r}_A}(t) = \begin{pmatrix}0\\0\end{pmatrix} + t\begin{pmatrix}4\\2\end{pmatrix}$

$\tilde{\mathbf{r}_A}(t) = \begin{pmatrix}4t\\2t\end{pmatrix} m$

Position vector vectors, at any time, t

$\tilde{\mathbf{r}_A}(t) = \begin{pmatrix}4t\\2t\end{pmatrix} m$ and $\tilde{\mathbf{r}_B}(t) = \begin{pmatrix}8\\8+2t\end{pmatrix} m$

$\overrightarrow{\mathbf{r}_{BA}}(t) = \overrightarrow{\mathbf{r}_B}(t) - \overrightarrow{\mathbf{r}_A}(t) = \begin{pmatrix}8\\8+2t\end{pmatrix} - \begin{pmatrix}4t\\2t\end{pmatrix}$

$\overrightarrow{\mathbf{r}_{BA}}(t) = \begin{pmatrix}8-4t\\8\end{pmatrix} \mathrm{m}$

$\overrightarrow{\mathbf{v}_{BA}} = \begin{pmatrix}0\\2\end{pmatrix} - \begin{pmatrix}4\\2\end{pmatrix} = \begin{pmatrix}-4\\0\end{pmatrix} ms^{-1}$

At closest approach, $\overrightarrow{\mathbf{r}_{BA}} \cdot \overrightarrow{\mathbf{v}_{BA}} = 0$,

$\begin{pmatrix}-4\\0\end{pmatrix} \bullet \begin{pmatrix}8-4t\\8\end{pmatrix} = 0,$

$-4(8-4t) = 0 \quad \Rightarrow \quad t = 2\,\mathrm{s},$

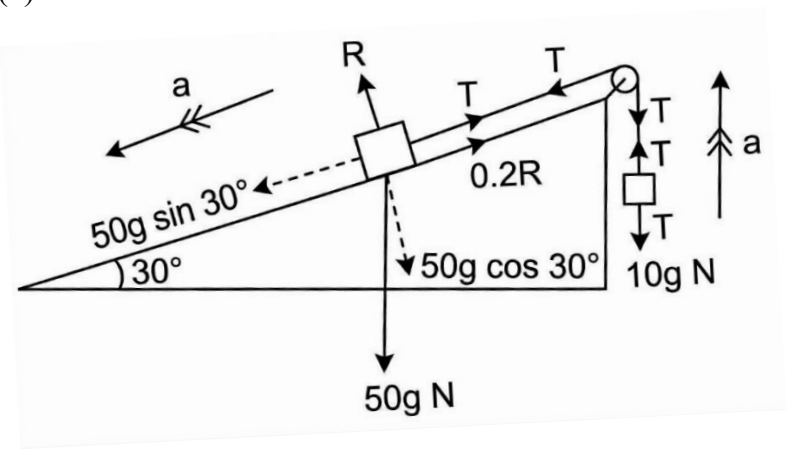
$\mathbf{r}_{BA}(t=2) = \begin{pmatrix}0\\8\end{pmatrix} \mathrm{m}.$

$d_{\min} = \sqrt{0^2 + 8^2}$

$d_{\min} = 8\,\mathrm{m}$

Thus $d_{\min} = 8\,\text{m}$, meeting the stated safety requirement.

(b) Connected bodies on an incline



For the hanging 10 kg mass
$T - 10g = 10a$,
$T = 10g + 10a$.
For the 50 kg mass
$R = 50g\cos 30°$
$50g\sin 30° - (T + 0.2R) = 50a$.
Substituting T and R gives

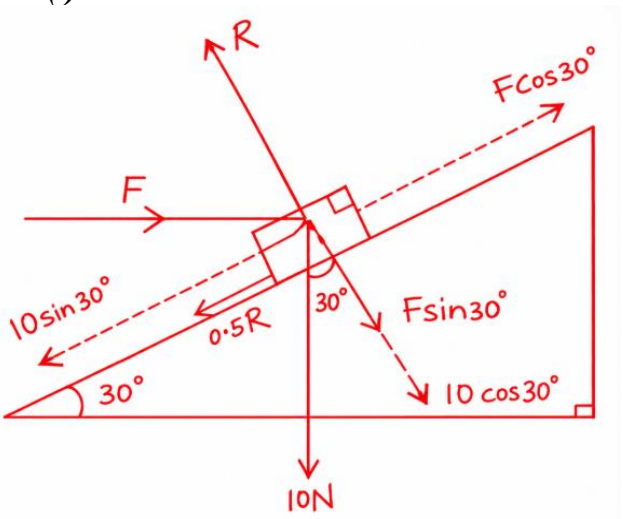
$$a = \frac{50g\sin 30° - 10g - 0.2(50g\cos 30°)}{60} = 1.0355\,\text{m s}^{-2}.$$

From rest over $s = 4\,\text{m}$,
$v^2 = 2as = 2(1.0355)(4)$,
$v = 2.8782\,\text{m s}^{-1} < 3.5\,\text{m s}^{-1}$.
The crate does not cause structural damage.

(c) Projectile from an incinerator
For launch speed $8\,\text{m s}^{-1}$ at 30°,
$y = 8t\sin 30° - \frac{1}{2}(9.8)t^2$
$\text{y} = 4t - 4.9t^2$.
At $y = 0.8\,\text{m}$,
$0.8 = 4t - 4.9t^2$
$4.9t^2 - 4t + 0.8 = 0$.
$t = 0.3504$ s and 0.4659 s,
$x = 8t\cos 30°$,
$x = 8(0.3504)\cos 30°$
$x = 2.4276\,m$
$x = 8(2.4276)\cos 30°$
$x = 3.2278$ m
Both lie outside the specified intake interval; the incinerator must be repositioned.

Item Three

Part (i)



Resolve perpendicular to the plane

$R = F\ sin\ 30^0 + 10\ cos\ 30^0$

Resolve parallel to the plane

$F\ cos\ 300 = 10sin300 + 0.5R$

$F\ cos\ 300 = 10sin300 + 0.5(F\ sin\ 300 + 10\ cos\ 300)$

$F\ cos\ 300 = 10sin300 + 0.5\ F\ sin\ 300 + 5\ cos\ 300$

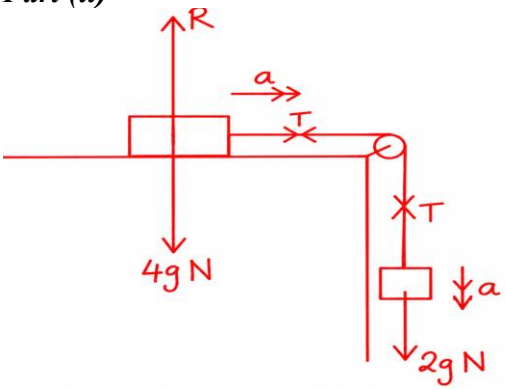
$F\ cos\ 300 - 0.5\ F\ sin\ 300 = 10sin300 + 5\ cos\ 300$

$$F = \frac{10 \sin 30° + 5 \cos 30°}{\cos 30° - 0.5 \sin 30°}$$

F = 15.1457N $<$ 30 N

Decision: The stated requirement cannot be met together with the condition that the container is exactly on the verge of sliding up the ramp.

Part (ii)



For the 2kg mass:

$2g - T = 2a$ ..... (i)

For the 4kg mass:

$T = 4a$ ..... (ii)

Eqn (i) + Eqn (ii)

$2g - T + T = 2a + 4a$

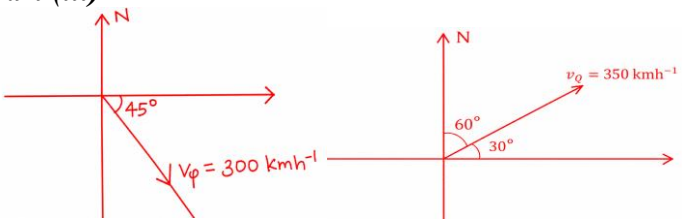
$2g = 6a$

$a = \frac{2\ g}{6}$

$a = \frac{2\ x\ 9.8}{6}$

$a = 3.2667\ ms^{-2} > 3.0$ m s^{-2}.
From (2)
$T = 4\ x\ 3.2667$
$T = 13.0668\ N < 15\ N.$
The loading operation should not proceed under the stated safety specification.

Part (iii)



Aircraft P

$\overrightarrow{V_p} = \begin{pmatrix} 300\cos 45^0 \\ -300\ sin45^0 \end{pmatrix} kmh^{-1}$

Aircraft Q

$\vec{V}_q = \begin{pmatrix} 350\cos 30^0 \\ 350\ sin30^0 \end{pmatrix} kmh^{-1}$

$\overrightarrow{V_{PQ}} = \overrightarrow{V_p} - \overrightarrow{V_q}$

$\overrightarrow{V_{PQ}} = \begin{pmatrix} 300\cos 45^0 \\ -300\ sin45^0 \end{pmatrix} - \begin{pmatrix} 350\cos 30^0 \\ 350\ sin30^0 \end{pmatrix}$

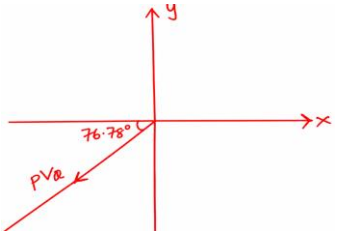
$\overrightarrow{V_{PQ}} = \begin{pmatrix} -90.9769 \\ -387.1320 \end{pmatrix} kmh^{-1}$

$\left|\overrightarrow{V_{PQ}}\right| = \sqrt{(-90.9769)^2 + (-387.1320)^2}$

$\left|\overrightarrow{V_{PQ}}\right| = 397.6782\ kmh^{-1}$

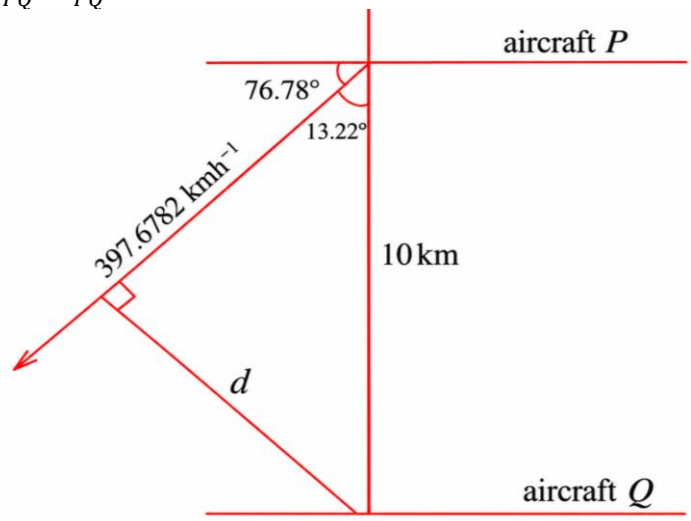
$\theta = \tan^{-1}(\frac{387.1320}{90.9769})$

$\theta = 76.78°$



For shortest distance

$\overrightarrow{V_{PQ}} . \overrightarrow{R_{PQ}} = 0$



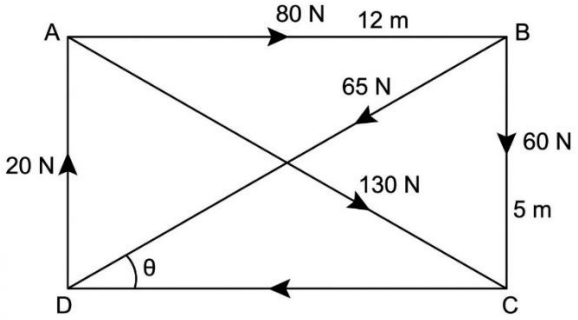
$\sin 13.22^0 = \frac{d}{10}$

$d = 10 \sin 13.22^0$

d = 2.2869 *km*

The minimum horizontal separation is approximately 2.2869 km, which is less than the required 5 km. Therefore the stated flight paths do NOT maintain the required separation throughout the flight. A change to the flight paths, speeds, bearings, or timing would be required for compliance.

Item Four



$\tan\theta = \frac{5}{12}$

$\sin\theta = \frac{5}{\sqrt{12^2+5^2}} = \frac{5}{13}$

$\cos\theta = \frac{12}{13}$

Let the resultant force be $R = \begin{pmatrix} Fx \\ Fy \end{pmatrix}$

$\begin{pmatrix} Fx \\ Fy \end{pmatrix} = \begin{pmatrix} 80 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -60 \end{pmatrix} + \begin{pmatrix} -40 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 20 \end{pmatrix} + \begin{pmatrix} -6\cos\theta \\ -6\sin\theta \end{pmatrix} + \begin{pmatrix} 130\cos\theta \\ -130\sin\theta \end{pmatrix}$

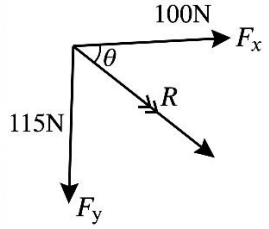
$\begin{pmatrix} Fx \\ Fy \end{pmatrix} = \begin{pmatrix} 40 \\ -40 \end{pmatrix} + \begin{pmatrix} -65(\frac{12}{13}) \\ -65(\frac{5}{13}) \end{pmatrix} + \begin{pmatrix} 130(\frac{12}{13}) \\ -130(\frac{5}{13}) \end{pmatrix}$

$\begin{pmatrix} Fx \\ Fy \end{pmatrix} = \begin{pmatrix} 100 \\ -115 \end{pmatrix} N$

$\overrightarrow{RF} = \sqrt{Fx^2 + Fy^2}$

$\overrightarrow{RF} = \sqrt{100^2 + (-115)^2}$

$\overrightarrow{RF} = \ 152.3975\ N < 155N$

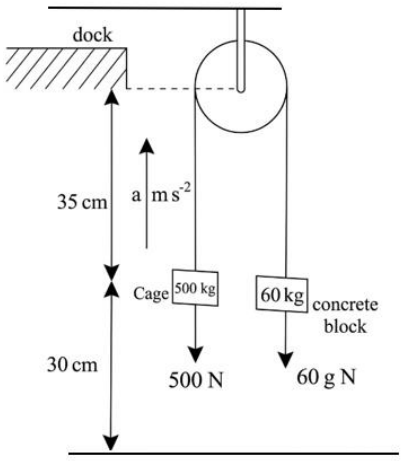


$\theta = \ tan^{-1}(\frac{115}{100})$

$\theta = \ 48.99^0$ which is in the range of $45^0 < \theta < 50^0$

: . they need to deploy the emergency anchor.

b)



<u>acceleration of the 500N block</u>
consider the 500N block
from $F = ma$

$$T - 500 = \left(\frac{500}{g}\right)a$$

Considering the 60kg block

$$60g - T = 60a$$

Solving

$$T - 500 = \left(\frac{500}{g}\right)a$$

$$T = \left(\frac{500}{g}\right)a + 500$$

Substituting

$$60g - \left[\left(\frac{500}{g}\right)a + 500\right] = 60a$$

$$60g - \frac{500a}{g} - 500 = 60a$$

$$60g^2 - 500a - 500g = 60ag$$

$$862.4 = 1088a$$

$$a = 0.7926\ ms^{-2}$$

Finding velocity of the cage after 30cm

$S = 30\ cm$, $a = 0.7926\ ms^{-2}$, $S = 0.3\ m$, $u = 0\ ms^{-1}$

from $v^2 = u^2 + 2as$

$$v^2 = 0 + 2as$$

$$v = \sqrt{2as}$$

$$v = \sqrt{2 \times 0.7926 \times 0.3}$$

$$v = 0.6896\ ms^{-1}$$

vertical distance remaining to the dock

$$s = (35 - 30)cm$$

$$s = 5\ cm$$

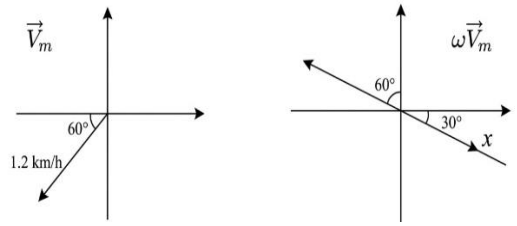
vertical distance the cage covered upwards

$u = 0.6896\ ms^{-1}$ $a = -9.8\ ms^{-2}$ $v = 0\ ms^{-1}$

$v^2 = u^2 - 2gs$
$u^2 = 2gs$
$s = \dfrac{u^2}{2g}$
$s = \dfrac{0.6896^2}{2 \times 9.8}$
$s = 0.03518m$
$s = 3.518$ cm < 5cm
:. The upward moving cage will not collide with and damage the rigid steal docking point.

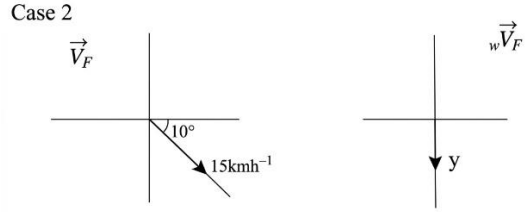
Part c

Case 1



$\overrightarrow{wVm} = \overrightarrow{Vw} - \overrightarrow{Vm}$
$\overrightarrow{Vw} = \overrightarrow{Vm} + \overrightarrow{wVm}$
Let $\overrightarrow{wVm} = x$
$\overrightarrow{Vw} = \begin{pmatrix} x\,cos\,30 \\ -x\sin 30 \end{pmatrix} + \begin{pmatrix} -12\cos 60 \\ -12\sin 60 \end{pmatrix}$
$\overrightarrow{Vw} = \begin{pmatrix} \frac{\sqrt{3}}{2}x - 6 \\ -0.5x - 6\sqrt{3} \end{pmatrix} kmh^{-1}$

Case 2

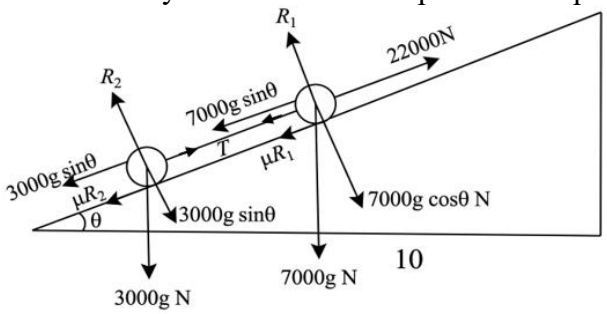


$\overrightarrow{Vw} = \overrightarrow{V_f} + \overrightarrow{wV_f}$
$\overrightarrow{Vw} = \begin{pmatrix} 0 \\ -y \end{pmatrix} + \begin{pmatrix} 15\cos 10 \\ -15\sin 10 \end{pmatrix}$
$\overrightarrow{Vw} = \begin{pmatrix} 14.7721 \\ -y - 2.6047 \end{pmatrix} kmh^{-1}$
For x-components
$\frac{\sqrt{3}}{2}x - 6 = 14.7721$
$x = 23.9856$
$\overrightarrow{Vw} = \begin{pmatrix} \frac{\sqrt{3}}{2}(23.9856) - 6 \\ -0.5(23.9856) - 6\sqrt{3} \end{pmatrix}$
$\overrightarrow{Vw} = \begin{pmatrix} 14.7721 \\ -22.3851 \end{pmatrix} kmh^{-1}$

$|\overrightarrow{Vw}| = \sqrt{(14.7721)^2 + (-22.3851)^2}$
$|\overrightarrow{Vw}| = 26.8199\ kmh^{-1} > 25\ kmh^{-1}$
Direction
$\theta = tan^{-1}(\frac{22.3851}{14.7721})$
$\theta = 56.58^0$
:. The fishing vessels must immediately return to the landing site.

Item Five

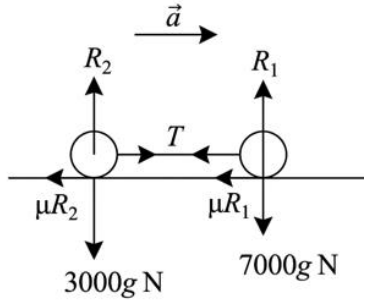
Final velocity of the car at the top of the ramp.



$sin\ \theta = \frac{1}{\sqrt{1^2 + 10^2}}$
$sin\ \theta = \frac{1}{\sqrt{101}}$
$cos\ \theta = \frac{10}{\sqrt{1^2 + 10^2}}$
$cos\ \theta = \frac{10}{\sqrt{101}}$
For 3000 kg mass
Resolving perpendicular to the plane
$R_2 = 3000gcos\theta$
Resolving parallel to the plane
$T - 3000gsin\theta - \mu R_2 = 3000a$
$T - 3000gsin\theta - 0.06(3000gos\theta) = 3000a$
$T - 3000g(sin\theta + 0.06cos\theta) = 3000a$ ……..1
Resolving perpendicular to the plane
$R_1 = 7000gcos\theta$
Resolving parallel to the plane
$22000 - T - 7000gsin\theta - \mu R_1 = 7000a$
$22000 - T - 7000gsin\theta - 0.06(7000gcos\theta) = 7000a$
$22000 - T - 7000g(sin\theta + 0.06cos\theta) = 7000a$ ……..2
Adding 1 and 2
$22000 - 10000g(sin\theta + 0.06cos\theta) = 10000a$
$a = \frac{22000 - 10000g\left(\frac{1}{\sqrt{101}} + 0.06\frac{10}{\sqrt{101}}\right)}{10000}$
$a = 0.6398\ ms^{-2}$
$U = 12ms^{-1}\ \ a = 0.6398ms^{-2}, s = 80\ m$
$v^2 = u^2 + 2as$

$v = \sqrt{12^2 + 2 \times 0.6398 \times 80}$
$v = 15.6961ms^{-1}$

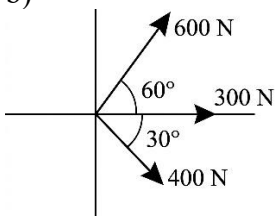
Distance to be covered horizontally



For 3000kg mass
Resolving vertically
$R_2 = 3000gN$
Resolving horizontally
$T - \mu R_2 = 3000a$
$T - 0.06(3000g) = 3000a \ldots\ldots\ldots\ldots (i)$
For the 7000kg mass
Resolving vertically
$R_1 = 7000gN$
Resolving horizontally
$-(T + \mu R_1) = 7000a$
$-T - 0.06(7000g) = 7000a \ldots\ldots\ldots\ldots (ii)$
$(i) + (ii)$
$-0.06 \times 10000 \times 9.8 = 10000a$
$a = -0.588ms^{-2}$

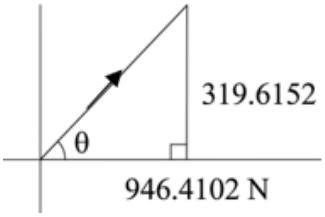
$V = 0ms^{-1},\ u = 15.6961\, ms^{-1},\ a = -0.588\, ms^{-2}$
$v^2 = u^2 + 2as$
$s = \dfrac{v^2 - u^2}{2a}$
$s = \dfrac{0^2 - 15.6961^2}{2 \times -0.588}$
$s = 209.4962m < 300m$
:. The heavy convoy safely stops before reaching the refueling station without applying manual brakes.

b)



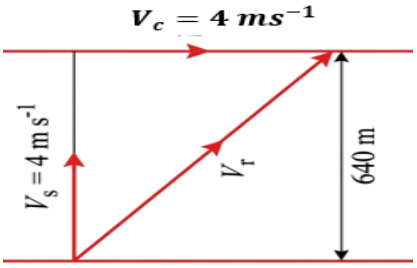
Resultant force $\boldsymbol{R} = \begin{pmatrix} 600\cos 60^0 \\ 600\sin 60^0 \end{pmatrix} + \begin{pmatrix} 300 \\ 0 \end{pmatrix} + \begin{pmatrix} 400\cos 30^0 \\ -400\sin 30^0 \end{pmatrix}$
$R = \begin{pmatrix} 946.4102 \\ 319.6152 \end{pmatrix} N$
$|R| = \sqrt{(946.4102)^2 + (-319.6152)^2}$

$|R| = 998.9224N < 1010N$



$\tan\theta = \frac{319.6152}{946.4102}$
$\theta = 18.66^0$
$90 - \theta = 90 - 18.66^0$
$90 - \theta = 71.34^0$
Direction is 071.34^0
$060^0 < 71.34^0 < 075^0$
:. The trailer's structural hitch remains the same during the alignment process.

Part (c)

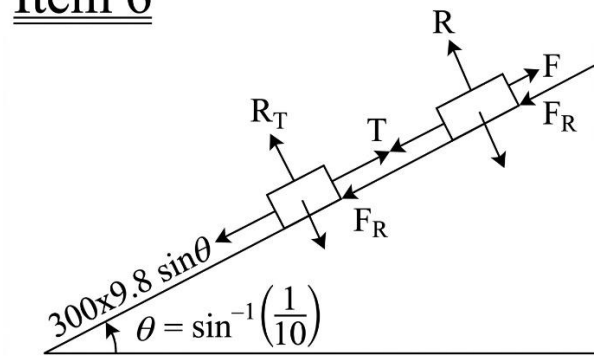


Time taken, $t = \frac{d}{Vs}$
$t = \frac{640}{4}$
$t = 160s$
$t = \frac{160}{60}$ minutes

$$t = 2.6667\ minutes < 3\ minutes$$

:. the men crossing did not cause the operational delay bottleneck

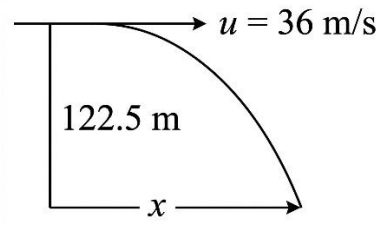
Item 6



$\sin\theta = \frac{1}{10}$, $t = \frac{d}{s}$
$t = \frac{450}{1.5}$

$t = 300s$
$t = 5\ minutes$
Resistance by truck
$F_R = 0.2 \times 300$
$F_R = 60N$
$T = \left(300 \times 9.8 \times \frac{1}{10}\right) + 60$
$T = 354.0N < 400N$
The trailer should be cleared for direct loading

b)



$t = \sqrt{\frac{2h}{g}}$
$t = \sqrt{\frac{2 \times 122.5}{9.8}}$
$t = 5s$
$x = ut$
$x = 36 \times 5$
$x = 180m$
Since $180\ m$ falls in the interval 175m < x < 185m, there will be successful delivery.

$v_x = 36\ ms^{-1}$
$v_y = 9.8 \times 5 = 49\ ms^{-1}$
$v^2 = u^2 + 2gs$
$v = \sqrt{36^2 + 49^2}$
$v = 60.8030\ ms^{-1}$
$v = 60.8030\ ms^{-1} > 60\ ms^{-1}$
:. The fragile components are not protected from structural failure upon impact.

c).
$\overrightarrow{R_{AB}}(t) = \overrightarrow{R_{AB}} + t\overrightarrow{V_{AB}}$
$\overrightarrow{R_{AB}} = \overrightarrow{R_A} - \overrightarrow{R_B}$
$= \begin{pmatrix}18\\6\end{pmatrix} - \begin{pmatrix}4\\10\end{pmatrix}$
$= \begin{pmatrix}14\\-4\end{pmatrix} km$
$\overrightarrow{V_{AB}} = \overrightarrow{V_A} - \overrightarrow{V_B}$
$= \begin{pmatrix}1\\4\end{pmatrix} - \begin{pmatrix}3\\2\end{pmatrix}$
$= \begin{pmatrix}-2\\2\end{pmatrix} kmh^{-1}$

$R_{AB}(t) = \begin{pmatrix} 14 \\ -4 \end{pmatrix} + \begin{pmatrix} -2 \\ 2 \end{pmatrix} t$

$R_{AB}(t) = \begin{pmatrix} 14 - 2t \\ -4 + 2t \end{pmatrix} km$

$\left|\overrightarrow{R_{AB}}(t) \cdot \overrightarrow{V_{AB}}\right| = 0$

$\begin{pmatrix} 14 - 2t \\ -4 + 2t \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 2 \end{pmatrix} = 0$

$-28 + 4t - 8 + 4t = 0$

$8t = 36$

$t = 4.5\ hours$

$d = \left|\overrightarrow{R_{AB}}\right|$

$d = \left|\begin{pmatrix} 14 - 2(4.5) \\ -4 + 2(4.5) \end{pmatrix}\right|$

$d = \left|\begin{pmatrix} 5 \\ 5 \end{pmatrix}\right|$

$= \sqrt{5^2 + 5^2}$

$d = 7.0712\ km > 5\ km$

:. The port controller must intervene with an emergency rerouting command.

END